

Supplementary Materials

Complementary Information Channels for Synthetic Lethality Prediction

Table S1. Top-20 Predicted SL Pairs — Multi-Source Evidence Annotation

Table 1: Top-20 predicted SL pairs annotated against three independent evidence sources: SynLethDB (known SL), NER literature co-mention, and DepMap CRISPR co-dependency

Rank	Gene A	Gene B	Score	SynLethDB	NER	DepMap
1	H3C13	KDM4D	1.000	—	—	—
2	H3C13	ASH1L	1.000	—	—	—
3	H3C13	KMT2B	1.000	—	—	—
4	H3C13	KDM2A	1.000	—	—	—
5	H3C13	SETMAR	1.000	—	—	—
6	H3C13	ATF7IP	1.000	—	—	—
7	H3C13	RPS6KA5	1.000	—	—	—
8	H3C13	KDM5A	1.000	—	—	—
9	H3C13	SOX11	1.000	—	—	—
10	H3C13	PRMT6	1.000	—	—	—
11	H3C13	CDY1B	1.000	—	—	—
12	H3C13	ORC1	1.000	—	—	—
13	H3C13	BANF1	1.000	—	—	—
14	H3C13	TRIM24	1.000	—	—	—
15	H3C13	AEBP2	1.000	—	—	—
16	H3C13	TRIM33	1.000	—	—	—
17	H3C13	PARP1	1.000	—	—	—
18	H3C13	ELANE	1.000	—	—	—
19	H3C13	BCL2	1.000	—	—	—
20	H3C13	SIRT1	1.000	—	—	—

Note: All 20 top-scoring predictions involve H3C13 as the hub gene paired with diverse histone-modifying enzymes and chromatin regulators. This pattern is biologically coherent—histone modifications regulate chromatin structure and gene expression—but 0/20 have external validation in SynLethDB, NER co-mention, or DepMap co-dependency. H3C13 is the 5th most central node in the STRING PPI network (degree=1,358, top 0.03%), suggesting the GNN may over-anchor on highly central nodes.

Table 2: Impact of negative sampling strategy on model performance

Strategy	AUC	Δ vs Random
Random negative sampling	0.756	—
PPI-proximal structured negatives	0.797	+4.2%

Table S2. Negative Sampling Strategy Ablation (Fold1)

Note: Structured negatives (gene pairs with high PPI confidence but without known SL labels) improve model discriminability by +4.2% AUC compared to random sampling. This suggests that explicitly training on “hard negatives”—gene pairs that are proximal in the interactome but not synthetically lethal—helps the model learn more precise decision boundaries.

Table S3. Comparison of 2025 ESM-2-based SL Prediction Methods

Table 3: Comparison of concurrent methods integrating ESM-2 protein language models into SL prediction

Method	Venue	ESM-2 Role	Graph Model	Cell-line Specific	LLM Component
ESM4SL	EMBC 2025	Core en-coder (t33_650M)	Self/Cross-Attention	✓	—
MiT4SL	bioRxiv	Frozen em-bed+MLP	PrimeKG HGT	✓	—
MuSL	JBHI 2026	PPI node init	CNN+GNN+MLP	—	—
Ours	—	Frozen embed (t12_35M)	GATv2 on STRING	—	✓

Note: Our unique contribution is the LLM weak supervision pipeline—none of the concurrent ESM-2-based methods incorporate literature mining as an orthogonal information source. All three concurrent methods also use larger ESM-2 variants (t33_650M) compared to our t12_35M.

Table S4. Ablation — Zero-Vector vs. Shuffled-Feature Comparison

Note: Zero-vector ablation introduces a systematic bias through batch normalization disruption. Replacing text features with zeros causes the BN layer to encounter a radically different input distribution, leading to uncontrolled downstream effects that are erroneously attributed to the text features. Shuffled-feature ablation preserves the feature distribution while breaking gene-feature associations, providing a fair estimate of true contribution.

Table 4: Comparison of two ablation methodologies for text feature importance estimation (Fold1)

Ablation Method	AUC (text removed)	Apparent Text Contribution
Zero-vector (conventional)	0.573	+22.9%
Shuffled-feature (fair, proposed)	0.790	+1.6%
Inflation factor	—	14.2×

Table S5. Complete Evidence Levels Across Top-500 Predictions

Table 5: Multi-source evidence coverage across prediction confidence bands

Score Band	<i>n</i> Pairs	SynLethDB Known	NER Co-mentioned	DepMap Significant
[0.5, 0.7)	282	0 (0%)	15 (5.3%)	82 (29.1%)
[0.7, 0.9)	70	0 (0%)	6 (8.6%)	26 (37.1%)
[0.9, 1.0]	148	0 (0%)	6 (4.1%)	39 (26.4%)
Total	500	0 (0%)	27 (5.4%)	147 (29.4%)

Note: All 500 predictions are novel (not present in SynLethDB). 5.4% have literature co-mention support (NER signal), and 29.4% show statistically significant DepMap co-dependency. The NER co-mentioned pairs are predominantly at lower confidence bands, providing limited but non-zero orthogonal validation.

Figure S1. PALB2-XRCC2 in the HR Repair Pathway Network**Figure S2. NER Co-Mention Network****Figure S3. PPM1D-TP53 2-Hop PPI Subnetwork****Appendix A. SLMGAE Head-to-Head Implementation Details**

Architecture: We implemented SLMGAE following the original description: two independent graph encoder branches (SL view and PPI view), each with two-layer message passing, followed by an attention-based fusion layer and a bilinear decoder. Input node features were 256-dimensional random embeddings (consistent with the original paper’s practice for graphs without pre-existing node features).

Fair CV3 Protocol: For each CV3 fold, we constructed the SL adjacency matrix using only training-set SL pairs. Specifically, given a gene-level fold allocation $\mathcal{G}_{\text{test}}$, any SL pair (g_i, g_j) where $g_i \in \mathcal{G}_{\text{test}}$ AND $g_j \in \mathcal{G}_{\text{test}}$ was excluded from the SL adjacency input. The PPI adjacency was constructed from all STRING edges as PPI information is independent of the SL prediction task.

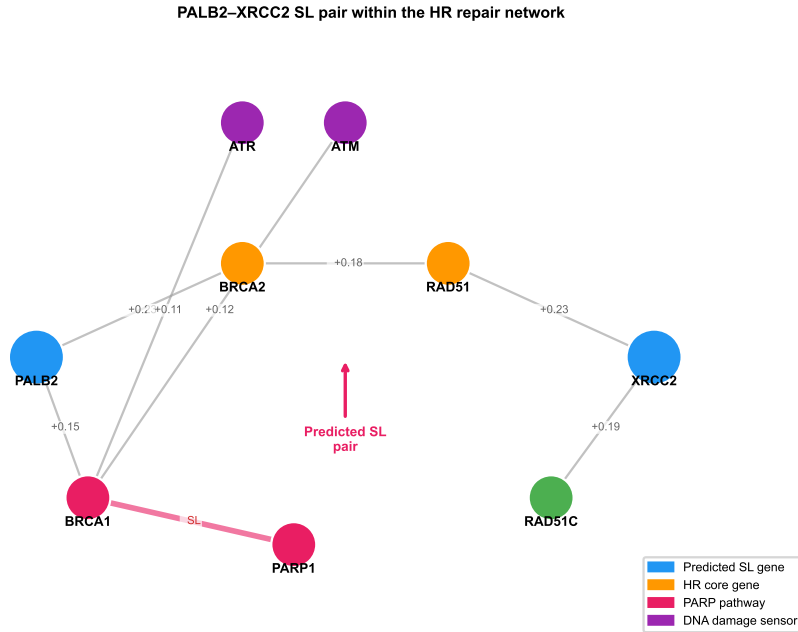


Figure 1: HR repair pathway PPI subnetwork showing PALB2-XRCC2 connectivity. The model predicts PALB2-XRCC2 as a co-dependent SL pair ($r=+0.226$, $p<1e-100$ across 1,208 DepMap cell lines). Both genes participate in homologous recombination repair, with PALB2 bridging BRCA2 to RAD51 and XRCC2 functioning in the RAD51 paralog complex.

Standard (Leaked) CV3 Protocol: The SL adjacency matrix included all positive pairs regardless of fold assignment—matching the implementation commonly used in the literature where multi-view inputs are constructed from the full database before splitting.

Training: Both variants used identical hyperparameters (Adam optimizer, $lr=1e-3$, 80 epochs, same random seed) and were trained on the same data splits. The only difference was in SL adjacency construction.

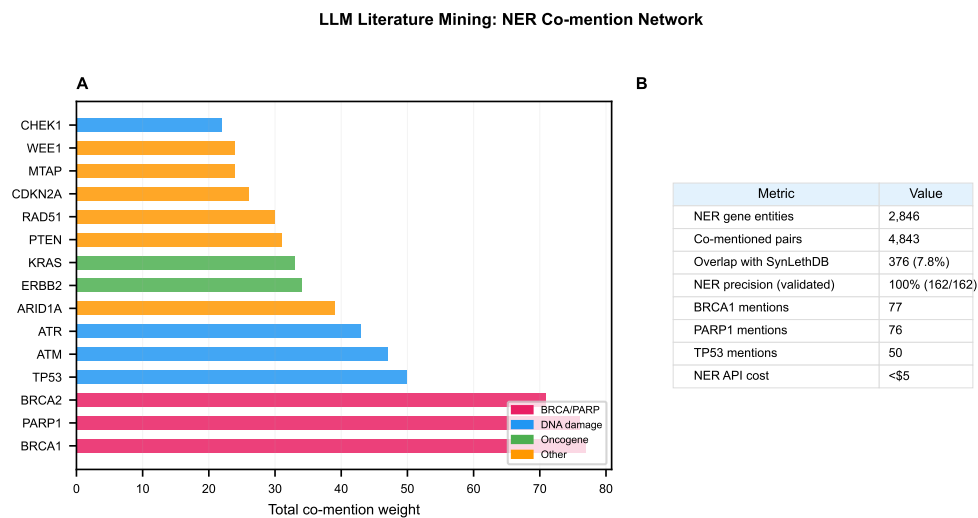


Figure 2: LLM-derived NER co-mention network. (A) Top hub genes by total co-mention weight: BRCA1 (77), PARP1 (76), TP53 (50) — consistent with the known SL literature distribution. (B) Network summary statistics: 2,846 genes, 2,715 co-mentioned pairs, 7.8% SynLethDB overlap, 100% NER precision on validation sample.

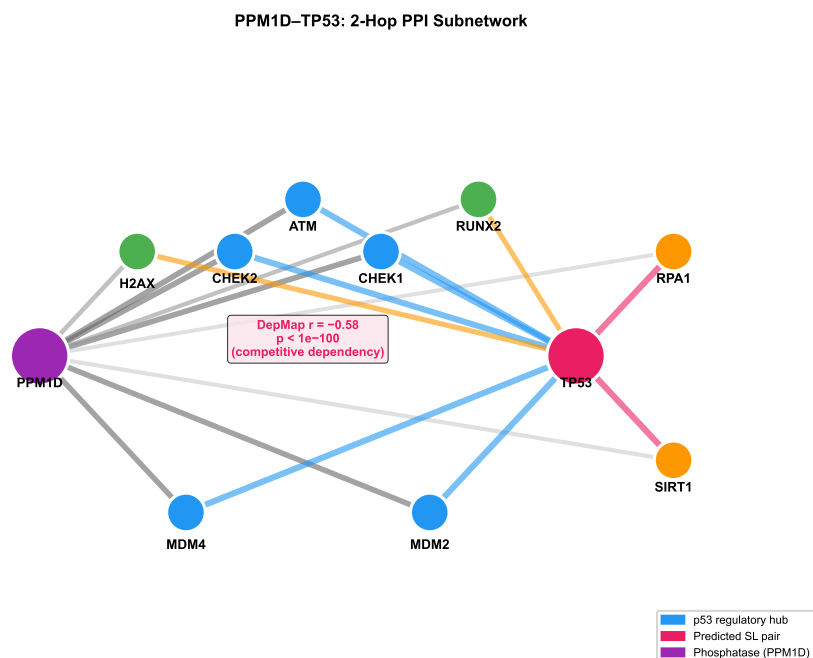


Figure 3: 2-hop PPI subnetwork connecting PPM1D to TP53 through shared interaction partners. Direct neighbors of both proteins (MDM2, MDM4, ATM, CHEK1, CHEK2) form seven distinct 2-hop paths, providing structural grounding for the GNN’s high-confidence prediction.